

CLONE WARS

Sixteen analogue clones of almost, but not quite, the same oscillator — drawn up in two armies of eight, aimed at each other, and slid into battle with one fader.



16 CLONES · 2 ARMIES · 3 NOTES

GROWL / SCREAM / LADDER

SPRING · TAPE · BBD · DRIVE

SERIAL CW-16 · BUILD 260825.1

WINDOWS VST3 + STANDALONE · FREE

01 A war between sixteen copies of one machine

Every great analogue drone has the same secret: many voices that are supposed to be identical, and are not. Sixteen oscillators built from the same schematic never agree — each one is a few cents off, a few degrees warmer, a little slower to wake. Clone Wars builds that disagreement on purpose.

The console holds **sixteen clones**, each a complete voice: an oscillator with four waveforms and four footages, an analogue-modelled filter, its own LFO, two looping envelopes, drift, pan and a fader. They stand in **two armies of eight** — Army A on the left, Army B on the right — with a command column between them. The two armies can carry two entirely different sounds, and **THE WAR** crossfade slides the mix from one to the other over anything from fifty milliseconds to five minutes.

What makes the machine breathe is the imperfection engine. **TOLERANCE** scatters component values across the sixteen clones, so no two strips are quite the same instrument. **DRIFT** lets every oscillator wander slowly around its pitch. **ENTRAIN** pulls the clones of an army gently into phase with each other, the way oscillators sharing a power rail fall into step. **TIDE** is a conductor moving over minutes, leaning on the whole orchestra at once. Between them, a held chord is never static: it swells, beats, locks and unlocks.

Everything on the panel is deterministic. A **seed number is a whole patch** — the same number is the same drone on any machine — and the panel morphs continuously between two seeds. Even the scratches are deterministic: the hull records its service hours and accumulates dents, chips and grime as it is played, and two numbers in your DAW project replay the whole damage history. There is no off switch for that. There is only **REPAIR**.

THE MACHINE IN ONE BREATH

16 clones	Complete voices — oscillator, filter, LFO, two looping envelopes, drift, pan, fader.
2 armies	Eight clones each, with their own filter temper, latch and base pitch — one crossfade between them.
3 notes	The machine listens to up to three held MIDI notes; every clone picks which one it follows.
1 hull	Ages with your project. The scars are yours.

02 Thirty seconds of instructions

A fresh instance is silent — a VST3 has manners. Give it a note and it becomes a monster.

DO THIS

WHAT HAPPENS

Play and hold a note

All sixteen clones wake on it. Hold two or three notes and the armies spread across them — each clone follows the held note its **NOTE 1·2·3** buttons point at.

Press KEYS

The field keyboard floats up into the keyboard bay under the console. Drag across it for glissandi; the computer keyboard plays too (A W S E D... is an octave).

Engage HOLD

Notes latch — release everything and the drone keeps standing. That is the drone way. HOLD drives LATCH A and LATCH B together.

Press PATCHES

Five families of seed patches — **abyss**, **swarm**, **engines**, **cathedral**, **rust**. Every number is the same sound on every machine.

Turn MORPH

The panel glides between seed A and seed B — two red LED numbers, one knob, a continuous journey between two complete patches.

Drag THE WAR

The mix marches from Army A to Army B. With SLEW up, the march takes minutes — set it moving and go make coffee.

Click GROWL / SCREAM / LADDER

Each army's filter temper. Same patch, three different animals.

Shift-click two knobs in a row

They lock — dragging one drags the whole locked group. The  dice reroll a row; the  fan ramps it across clones 1 → 16.

If nothing sounds: the machine is idle by design until it is played. Check a note is held (the HELD N1·N2·N3 lamps in the command column light for each occupied note slot), the strip faders are up, and no MUTE is engaged. The host parameter **drone** can wake it permanently with no input at all — see chapter 06.



03 Two minutes, no installer

A VST3 plugin is a folder you copy into place. That is the whole ceremony.

STEP ACTION

- 1 Unzip `Clone-Wars-VST3-win64.zip` (right-click → *Extract All*). Inside: the `Clone Wars.vst3` folder, the standalone `Clone Wars.exe`, this manual, and a readme.
 - 2 Copy the whole `Clone Wars.vst3` folder — not just a file from inside it — to `C:\Program Files\Common Files\VST3\`. A sub-folder such as `...\VST3\Brokild\` is fine; hosts look inside. Windows asks for administrator permission; that is normal.
 - 3 Rescan plugins in your DAW (Ableton Live: *Preferences* → *Plug-Ins* → *Rescan*). Clone Wars appears as an **instrument**, by Brokild. Put it on a MIDI track and hold a chord.
- or Skip the DAW: run `Clone Wars.exe`. It picks its audio and MIDI devices from its own options menu and has the field keyboard built in.

If the plugin window is blank or shows an error page: install the free **Microsoft Edge WebView2 runtime** from developer.microsoft.com/microsoft-edge/webview2 and reopen the window. The interface is drawn with WebView2, which every current Windows 10 and 11 ships with, but which can be missing on older or heavily stripped installations. The audio engine runs either way.

If Windows refuses to load the plugin and mentions an *Application Control policy*, that is **Smart App Control**. It judges each file from a cloud reputation lookup and can block one unsigned binary while loading another happily. Most machines do not have it enabled at all; Windows Security → App & browser control shows whether yours does.

REQUIREMENTS

System	Windows 10 or 11, 64-bit
Format	VST3 instrument, stereo out, MIDI in — plus a standalone application in the same zip
Sample rates	Any. Tested at 44.1, 48, 88.2 and 96 kHz.
CPU	One core carries the full sixteen-clone console with headroom at HQ; LOW and XHQ tiers are a click away (chapter 10)
Price	Free. No installer, no account, no telemetry, no network access at all.

04 Anatomy of a clone

Each of the sixteen strips is the same soldier, read top to bottom. The row labels stand on both wings of the console, so nothing is ever more than a glance away.

ROW	WHAT IT DOES
LEVEL VU	A real needle VU with real ballistics, one per clone — sixteen needles leaning together is half the show.
NOTE 1·2·3	Which of the three held MIDI notes this clone follows (chapter 06).
WAVE	SAW / PULSE / TRI / SINE — band-limited analogue-modelled cores.
FOOTAGE	32' / 16' / 8' / 4' — organ-style octave. The 32' rank is the floor-shaker; footages 16' and below can be pulled to mono centre on the bus (chapter 09).
TUNE	Microtune, a few cents either way. Where the beating between clones lives. The engine-room SPREAD knob fans all sixteen at once.
CUTOFF · RES	The clone's own analogue-modelled filter — its circuit is chosen per army by the TEMPER switch (chapter 05).
LFO WAVE · RATE	SIN / TRI / SQR / S&H per clone, with its own rate — sixteen LFOs at sixteen speeds is what makes the field shimmer.
LFO → AMP · LFO → FLT	How much of that LFO reaches the clone's level and its filter cutoff.
ENV FLT · ENV AMP	Two one-knob envelopes. The knob is a journey: hard left is fast attack / fast decay (percussive), noon is fast / slow (struck, then breathing), hard right is slow / slow (tidal). One knob, the whole family of shapes.
ENV LOOP	The envelope self-retriggers — the strip breathes on its own, its period set by the envelope travel.
DRIFT	How far this clone wanders around its pitch, scaled by the command column's DRIFT master.
PAN · FADER	Position and level in the army's mix.
M · S	Mute and solo, per clone.

ROW WARFARE — COMMANDING SIXTEEN STRIPS AT ONCE

Shift-click	Locks any combination of controls in one row into a group — drag one and the whole group moves. Lock four LFO RATE knobs and sweep them together; the lock ring marks the members.
☒ dice	Rerolls a whole row from the seed — locked controls roll together.
↪ fan	Draws a ramp across clones 1 → 16 — a filter sunrise, a pan sweep, a stagger of envelope speeds, in one click.
WAR PAINT	Very faint colour rings sort the rows into families — oscillator, filter, LFO, envelope, mix. The ring of a row lights a little brighter while you touch it, then fades back into the paint.

05 The command column

Everything that rules the whole machine stands in the centre, between the armies. Top to bottom:

STATION	FUNCTION
SEED • MORPH	Two red LED seed numbers — A and B — and a MORPH knob that glides the whole console continuously between them (chapter 07).
PATCHES • RANDOM	The seed browser, five families deep — and a reroll of everything from a fresh seed.
ENGINE	Cycles the render tier: LOW / HQ / XHQ (chapter 10).
TOLERANCE	Component scatter. At zero the sixteen clones are one perfect machine; opened up, every strip is a slightly different instrument — filter, envelope and LFO all wander from spec, deterministically, from the unit's own serial seed.
TIDE	The conductor. A random walk moving over minutes that leans on the whole orchestra — level, cutoff, pitch — the way a drone piece swells without anyone touching anything.
ENTRAIN A • B	Phase pull inside each army. Low, the clones beat freely against each other; high, they fall into step like oscillators sharing a rail — the beating slows, thickens, and locks.
DRIFT	Master depth for every clone's pitch wander.
TEMPER A • B	Each army's filter circuit: GROWL — a Sallen-Key voice with diode squash, gnarly and bass-keeping; SCREAM — the same circuit pushed far harder, howling on its own harmonics near the top; LADDER — a four-pole transistor ladder, round, fat, Moog-mannered. One switch turns an army into a different animal.
LATCH A • B	Per-army note latch — released notes keep sounding for that army. The keyboard's HOLD switch drives both at once.
HELD N1•N2•N3	Lamps for the three note slots — lit while a slot is occupied.
THE WAR	The A ↔ B crossfade, equal-power, with SLEW — the fader's travel time, from fifty milliseconds to five minutes — and the MASTER level beside it (chapter 08).
BACK PANEL	The effects: SPRING (dwell, mix, and FREEZE — an infinite spring for drone sustain), BBD bucket-brigade chorus (rate, depth), TAPE echo (60 ms–1.2 s, feedback, mix), and DRIVE for the whole bus.
ENGINE ROOM • WAR PAINT • KEYS	The under-the-hood hatch (chapter 09), the row colour rings, and the field keyboard (chapter 06).

The armies are two different sounds, not one sound twice. Give Army A a 32'-heavy LADDER abyss and Army B a screaming 4' swarm, park THE WAR in the middle, and the machine plays a duet with itself. Then set SLEW to minutes and let the war decide.

06 Three notes, sixteen clones, one keyboard

Most drones are one note. Clone Wars listens to three at once — and lets every clone choose which one it obeys.

The machine keeps **three note slots**, filled in order of arrival: the first held note takes slot 1, the second slot 2, the third slot 3. A fourth note steals the oldest slot. Each strip's **NOTE 1·2·3** buttons say which slot that clone follows — so a held three-note chord can be voiced any way you like across the armies: all of Army A on the bass note, Army B split across the fifth and the octave, one lone clone riding the top. When a clone's slot is empty it falls back to the lowest occupied one, so the drone never drops a soldier.

LATCH A and **LATCH B** hold notes after release, per army. With latch off the machine goes quiet when you let go, like an instrument; with latch on it stands where you left it, like a monument. Fresh instances open with latch off and make **no sound until played** — a VST3 has manners. For an installation drone that must sound with no input at all, automate the host parameter **drone**: it wakes the machine on its base pitch, no MIDI required.

THE FIELD KEYBOARD

Press **KEYS** and a three-octave field keyboard (C2–C5) floats up into the **keyboard bay** beneath the console — a two-second landing, drifting in from below the deck with its shadow growing on the hull, spaceship fashion. Press **KEYS** again and it departs the same way. The bay is its own deck: the keyboard covers no controls while deployed.

Mouse	Click to play, drag across for glissandi.
Computer keys	A W S E D F T G Y H U J K — home row plays an octave, black keys on the row above.
HOLD	The latch switch on the keyboard plate — engages LATCH A and B together, and mirrors them both ways.
Livery	The keys wear the hull's own paint: the black keys are panel blue, the whites a lighter blue of the same metal. Unusual for a keyboard. This is an unusual machine.

Voicing recipe: hold a low C, a G above it, and a high C. Set Army A's clones to NOTE 1, Army B's alternating 2 and 3. Now THE WAR is not just a timbre crossfade — it is a harmonic one, marching from the root to the interval and back.

07 Seeds, families, and the morph

A seed number **is** a patch. Not a pointer to one — the patch itself, generated from the number, bit-identical on every machine, forever.

Dial a seed and the entire console jumps to it: all sixteen strips, the tempers, the imperfection engine, the effects. Each seed belongs to a **family** chosen first, so the category and the sound always agree:

FAMILY	CHARACTER
ABYSS	32' and 16' footage, LADDER tempers, slow tides — the bottom of the ocean.
SWARM	High footages, sharp tolerances, fast scattered LFOs — a field of insects with opinions.
ENGINES	Pulse-heavy, entrained, mechanical — the hum of a hull under power.
CATHEDRAL	Wide, spring-frozen, slow envelopes — sacred space in a box.
RUST	Detuned, driven, S&H-modulated — beautiful decay.

The command column carries **two** seeds at once — the red LED numbers A and B — and the **MORPH** knob glides the whole console continuously between them. Every knob value travels; the selectors switch over at the halfway point. A slow morph between an ABYSS and a RUST is a composition in itself. **RANDOM** deals a fresh seed; **PATCHES** browses all of them by family.

A seed is a postcard. Tell a friend "CLONE WARS, seed 042" and they hold your exact drone. Tweak any knob afterwards and the sound is yours — the DAW project saves every parameter as it stands, and the seed display still shows where you started.

08 A crossfade with a strategy

THE WAR is the horizontal fader in the command column: hard left is Army A alone, hard right is Army B, equal-power all the way across. What makes it a weapon is **SLEW** — the fader's own travel time. At the bottom the war snaps in fifty milliseconds; at the top, a single drag takes **five minutes** to arrive, the two armies trading ground so slowly the change is felt before it is heard. Automate the target from the DAW and the slew still shapes the journey — the fader on the panel shows the war's true position, not the order it was given.

09 The engine room

The front panel is for playing. The **ENGINE ROOM** button opens the hatch to everything else — four bolts, one dark room.

STATION	FUNCTION
BASE A · BASE B	Each army's base pitch — where the clones sit when the drone parameter drives them, and the floor the note slots transpose from. Two armies, two keys.
GLIDE	Portamento between held notes, up to two seconds — the drone way of changing chords.
SPREAD	Fans all sixteen microtunes symmetrically — the one-knob version of detuning every strip by hand.
KBD A · KBD B	Filter key-tracking per army — how much the cutoff follows the note.
SCATTER · LFO SYNC	Panel fitted, wiring scheduled: SCATTER will re-randomise all sixteen LFO phases, LFO SYNC will lock the LFO rates to the host clock. In this build the LFO phases come from the seed and free-run.
WIDTH	Master stereo width, from mono through unity to wide.
BUS SAT	Analogue bus soft-clip with automatic gain compensation — glue at low settings, weld at high ones.
BASS MONO	Pulls footages 16' and below to centre — the low ranks stand firm while the high ranks fight across the stereo field.
HPF	A gentle infrasonic high-pass on the master — sixteen drifting oscillators can conspire below hearing; this keeps the conspiracy off your woofers.
MODULE BAY	Four empty slots. Future plug-in modules dock here — the bay is real, the modules are coming.

The output is always safe. Underneath everything sits a transparent soft ceiling: it does nothing at all until the mix approaches full scale, then rounds the peaks instead of clipping them. Nothing this machine can do — not sixteen screaming filters, not a frozen spring into full drive — puts a hard edge above full scale into your monitors.

10 LOW · HQ · XHQ

The ENGINE button cycles three render tiers. They change how hard the nonlinear stages — the filters and saturators, where aliasing is born — work per sample:

TIER	WHAT IT MEANS
LOW	No oversampling. The full console at minimum CPU — for sketching on a laptop with the fans off. Still band-limited oscillators, still the soft ceiling.
HQ	Nonlinearities run 2× oversampled. The default: transparent for everything short of the most violent SCREAM settings.
XHQ	4× oversampled nonlinearities — the measured aliasing floor of the whole engine. Costs real CPU with sixteen clones running.

Bounces are always XHQ. When the host renders offline — an export, a freeze, a bounce-in-place — the engine ignores the panel switch and runs XHQ. Play at LOW all day; the record is always made at full quality.

11 The hull remembers

Every instance of Clone Wars keeps an odometer — **UNIT AGE**, top right, in hours of service. While the unit plays, it ages. And as it ages, it scars.

Dings, chips, paint peel, rust breakthrough, smears and fingerprints accumulate on the panel with use — photographic damage, placed deterministically from the unit's own serial, so your scars are yours and stay exactly where they fell. The age and the damage history are saved **per instance, inside the DAW project**: the Clone Wars in your old project is old, the one you added today is factory fresh. Hard sessions at high level scar faster than quiet ones.

There is **no off switch** for any of this. There is only repair: **shift-click the odometer** and the **SERVICE BAY** hatch opens. **REPAIR** restores the hull to factory condition and starts the service clock again — new panel, day zero. **TOUR** does the opposite: it simulates a year of hard touring, right now, if you want an instrument with a past the day you meet it.

12 Specifications & ship's papers

SPECIFICATIONS

Instrument	16-voice analogue-modelled drone synthesizer — two 8-clone armies
Oscillators	16 × band-limited SAW / PULSE / TRI / SINE, footages 32'–4', per-clone microtune and drift, tolerance scatter, per-army entrainment
Filters	Per-clone analogue-modelled filter; per-army temper — GROWL and SCREAM (Sallen-Key with diode squash) or LADDER (four-pole transistor ladder with per-stage saturation)
Modulation	16 × LFO (SIN/TRI/SQR/S&H), 32 × one-knob looping envelopes, TIDE conductor, minutes-scale WAR slew
MIDI	Three note slots, arrival-ordered, per-clone slot select, per-army latch, glide
Effects	Spring reverb with FREEZE, tape echo, BBD chorus, bus drive; bass-mono, infrasonic HPF, width, auto-gained bus saturation, transparent output ceiling
Patches	Deterministic seed patches in five families, two seed slots with continuous morph
Quality	LOW / HQ / XHQ (1× / 2× / 4× oversampled nonlinearities); offline render always XHQ
Persistence	All parameters, both seeds, unit age, damage history and unit serial saved in the DAW project
Format	Windows 64-bit VST3 instrument + standalone · build 260825.1

SHIP'S PAPERS

CLONE WARS · Brokild Heavy Industries, Drone Division · Serial CW-16.

From the curved mind of **Professor Brokild**. Programmed with VS Code, ChatGPT and Claude. Native C++ audio engine, JUCE 8 shell, interface drawn in the plugin's own window. Source and issues:

github.com/peterboggild/BrokildApps.

Use of this instrument is at your own peril. It may produce unique, beautiful, terrifying or useless drones, and it may crash your DAW project. The hull damage is cosmetic; the machine inside is always factory-true. Play unsafely, unwisely and unboringly.

Free — no installer, no account, no telemetry, no network access at all. Cheers, Peter Bøggild, August 2026.

— END OF MANUAL · FIRST EDITION —